

A new technique for preventing control windup

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SUMMARY

In industrial control loops several situations can occur in which the output of a controller is different from the actual control signal applied to the process.

If no precautions are taken, serious disturbances in the controlled system can occur.

Producers of control equipment are well aware of this problem and are offering several practical solutions to it. But up to now there appears to be no general theory available covering the complete field of problems.

In this paper such a theory is developed.

Several applications to classical PID-controllers are given as an illustration of this theory.

1. INTRODUCTION

A classical industrial control loop is shown in figure 1. The loop consists of four elements : the controlled system S, the controller C, the measuring transmitter T and the actuator (or final controlling element) A. The output signal $y^m(t)$ of the measuring transmitter is compared with the set point $y^r(t)$; from the resulting error signal $e(t)$, the controller computes a control signal $u(t)$.

This signal is applied to the final controlling element (this could be a valve, a pump, a motor...), which transforms the low power control signal into a high power control signal $u^r(t)$ (we shall call u^r the "manipulated variable") applied to the controlled system.

The choice of the controller structure and parameters is made by assuming the control loop to be and to remain a linear system.

When, due to whatever cause, this assumption becomes false, the so-called windup phenomenon occurs.

The purpose of this paper is to analyze the windup phenomenon, to review the characteristics of a good anti-windup system and to propose an anti-windup system which competes very favourably with the ones described in the literature.

2. WINDUP PHENOMENON

The windup phenomenon occurs when the controlling signal $u(t)$ delivered by the controller is not in agreement with the output signal $u^r(t)$ of the actuator. The discrepancy between the two signals can cause a considerable degradation in the performance of the control loop, even long after the cause of the trouble

has vanished.

The origin of this discrepancy can be twofold. The first one is the substitution for the controlling signal $u(t)$ of another signal $u'(t)$ generated by another controlling device. Limitations of some kind or other (nonlinearities) of the output of the final controlling element are the second one.

Examples of the said substitution could be : the switching over to manual control (switch K_1 in fig. 1), the use of selective control, etc.

Frequently encountered limitations of the final controlling element are amplitude and rate constraints on its output.

In order to precise the meaning of the phenomenon, we will examine the P+I+D (proportional plus integral plus derivative) controller in this connection. The controlling signal $u(t)$ of a P+I+D controller is composed of three terms. The first, $P(t)$, is proportional to the error signal $e(t)$, the second $I(t)$ to its integral and the last $D(t)$ to its derivative :

$$u(t) = P(t) + I(t) + D(t) \quad (1)$$

$$P(t) = K_p e(t) \quad (2)$$

$$I(t) = I(t_0) + K_I \int_{t_0}^t e(\tau) d\tau \quad (3)$$

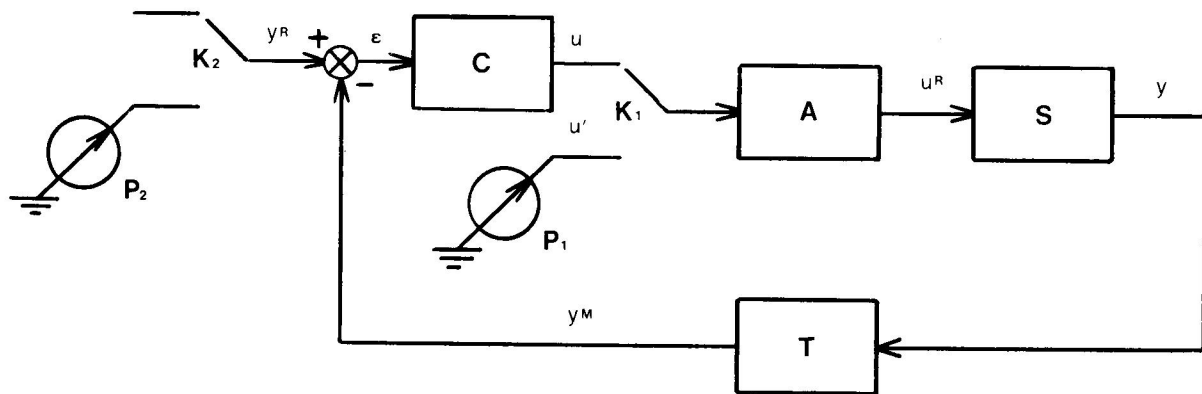
$$D(t) = K_D \frac{de(t)}{dt} \quad (4)$$

In the same way, when a digital computer is used to perform the control, it is possible to realize a similar control action by having recourse, among others, to one of the following two algorithms : the so-called position and velocity algorithms.

The equations of the position algorithm are :

$$u(k) = P(k) + I(k) + D(k) \quad (5)$$

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C : CONTROLLER ELEMENT
A : ACTUATOR
S : CONTROLLED SYSTEM
T : MEASURING TRANSMITTER

Fig. 1.

$$P(k) = K_P \cdot \epsilon(k) \quad (6)$$

$$I(k) = I(k_0) + K_I \sum_{\ell=k_0+1}^k \epsilon(k-\ell) \quad (7)$$

$$D(k) = K_D \{ \epsilon(k) - \epsilon(k-1) \} \quad (8)$$

with the important particular form of (7)

$$I(k) = I(k-1) + K_I \epsilon(k) \quad (9)$$

where the initial value k_0 is chosen equal to $k-1$.
The equations of the velocity algorithm are :

$$\Delta u(k) = \Delta P(k) + \Delta I(k) + \Delta D(k) \quad (10)$$

$$\Delta P(k) = K_P \{ \epsilon(k) - \epsilon(k-1) \} \quad (11)$$

$$\Delta I(k) = K_I \epsilon(k) \quad (12)$$

$$\Delta D(k) = K_D \{ \epsilon(k) - 2\epsilon(k-1) + \epsilon(k-2) \} \quad (13)$$

$$u(k) = u^r(k-1) + \Delta u(k) \quad (14)$$

This algorithm is particularly well adapted when the actuator, like a step motor for instance, presents a reset action.

However, with regard to the behaviour of the closed loop, the use of one or the other algorithm just amounts to the same thing when the a priori admitted linearity hypothesis is verified. The differences appear only in cases where controller windup takes place.

As we shall be able to see, the windup phenomenon is to be connected with the information storage properties of the controller.

As the memories are different in the two algorithms, they will react differently to the same situation.

In order to illustrate this, we have treated an example with a sampled system of the first order, controlled by different P+I algorithms.

The controlling signal $u(t)$ is rate constrained :

$$\Delta u_{\max} = 0,1 \quad (15)$$

The transfer function $F(s)$ of the controlled system is :

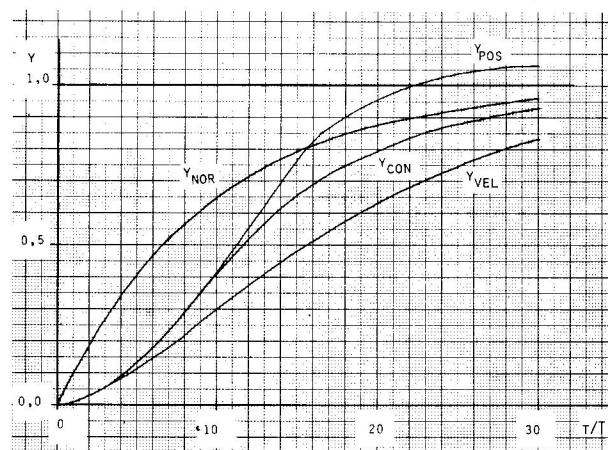


Fig. 2.

$$F(s) = \frac{1}{1+9,49 Ts} \quad (16)$$

where T is the sampling period and s the Laplace variable.

The transfer function $R(z)$ of the P+I controller is :

$$R(z) = \frac{1 - 0,9 z^{-1}}{1 - z^{-1}} \quad (17)$$

where z^{-1} is the delay operator.

The parameters of the P+I controller are chosen, in such a way that the zero of the controller compensates exactly the pole of the controlled system, and that the dynamics of the closed loop are identical with the dynamics of the controlled system. In this case, a stepwise variation of the set point signal generates a step of the controlling signal too.

In figure 2, different step responses are shown. The curve Y_{NOR} is the step response of the system described above, when there is no constraint on the manipulated variable.

The curve Y_{POS} is the step response of the system in the presence of the rate limit, when the position algorithm is used.

The increase of the integral term I in this instance causes an overshoot of the response.

The position algorithm tends generally to destabilize the system when a windup occurs.

The curve Y_{VEL} is the step response of the system using the velocity algorithm in the same instance.

The vanishing of the proportional term P causes sluggishness of the response.

The curve Y_{CON} shows the step response which can be obtained when the anti-windup algorithm we propose is used. The "conditioning" of the controller is then such that after the instant where saturation ceases, the step response is an exact translation of the response in the absence of any limitation. We can also see that the origin of the windup phenomenon lies in an incorrect storage of the state variables of the controller; the solution to this problem must be found in an adequate "piloting" of these state variables.

3. ANTI-WINDUP SYSTEMS

The anti-windup systems proposed in the literature [1] to [5] all lack generality. They are dependent of the control structure, of the controller type, of the nature of the constraints on the manipulated variable, etc.

These systems are complicated and moreover do not generally ensure good results.

A good anti-windup system should possess the following properties :

- independence of the control structure
- independence of the controller type
- independence of the controlled system
- independence of the nature of the constraints on the manipulated variable
- simplicity.

A general description of a one-input-one-output linear controller can be :

$$\dot{v} = A v + m \epsilon \quad (18)$$

$$u = b v + n \epsilon \quad (19)$$

where $v(t)$ is the state vector of the controller, $\epsilon(t)$ the scalar input signal (the control error signal) and $u(t)$ the scalar output signal (the controlling signal). We assume that the control obtained by this controller is satisfactory as long as no discrepancy between the control signals appears.

In the presence of a windup cause we decide to keep the same control structure as defined by (18) and (19), but to substitute for the actual error signal $\epsilon(t)$ another input signal $\epsilon^r(t)$ chosen in such a way that the resulting output signal $u(t)$ of the controller is equal to $u^r(t)$.

It is so hoped that the state of the memory part of the controller will be satisfactory since it coincides

with a state which would exist in the controller in a situation where no windup cause is present.

We also demand that the controller should again be described by (18) (19) as soon as the windup cause disappears.

So we are led to write the equations of the conditioned controller as follows :

$$\dot{v} = A v + m \epsilon^r \quad (20)$$

$$u^r = b v + n \epsilon^r \quad (21)$$

$$u = b v + n \epsilon \quad (19)$$

(19) (21) entail :

$$\epsilon^r = \epsilon + \frac{u^r - u}{n} \quad (22)$$

The set of equations (22), (20) and (19) constitutes the "conditioned" control algorithm we advocate.

It is to be pointed out that except for the scalar equation (22), the conditioned control algorithm is in no way more complicated than the original one. Figure 2bis is a block diagram of the conditioned controller.

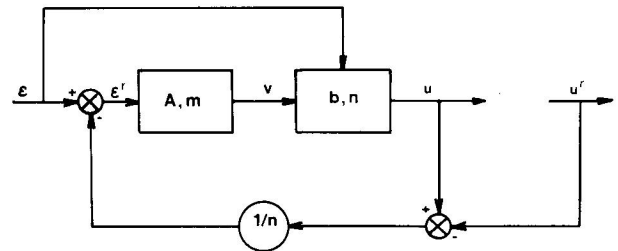


Fig. 2bis.

A transfer function description of the conditioned algorithm we propose can readily be obtained.

The transfer function of the unconditioned controller is, from (18) (19)

$$R(s) = \frac{U(s)}{E(s)} = b (Is - A)^{-1} m + n \quad (25)$$

where I is the identity matrix of the same order as A . We write

$$R_\infty \triangleq \lim_{s \rightarrow \infty} R(s) = n \quad (26)$$

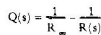
transform (20), (19), (22) and use (25) and (26) to get :

$$U = R \left\{ E - \left\{ \frac{1}{R_\infty} - \frac{1}{R} \right\} \{ U - U^r \} \right\} \quad (28)$$

The controlling signal $u(t)$ depends not only on the error signal $\epsilon(t)$ but also on the difference $u^r(t) - u(t)$. A block diagram corresponding to this result is given in figure 3.

The transfer function of the inner loop which appears on that block diagram is :

$$\frac{U(s)}{U^r(s)} = -\frac{R_\infty}{R(s)} + 1 \quad (29)$$



As the unconditioned controller is stable, the inner loop will also be stable if the unconditioned controller is a minimum phase system.

In order to illustrate the advantages of the proposed conditioning method, we give a very simple example (fig. 4a).

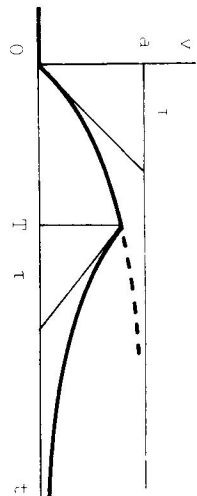
$$\mathbf{F}(\mathbf{s}) = \mathbf{1} \quad (30)$$
$$R(s) = K(1 + \frac{1}{s T_i}) \quad \text{with } T_i > 0 \quad (31)$$
$$u^r(t) = \begin{cases} -b & u(t) < -b \\ u(t) & -b \leq u(t) \leq b \\ +b & u(t) > b \end{cases} \quad \text{with } b > 0 \quad (32)$$
$$z(t) = \begin{cases} 0 & t < 0 \\ a & 0 \leq t < T \\ 0 & t \geq T \end{cases} \quad \text{with } a > 0, T > 0 \quad (33)$$
$$K > \frac{1}{\epsilon} \quad (34)$$

$$\frac{K}{K+1} \frac{T}{T_i} > \frac{1}{\frac{a}{b} - 1} \quad (35)$$

When no precaution is taken to prevent the windup phenomenon, the state variable $v(t)$ (the integral term $I(t)$ in this case) increases linearly because the error signal $e(t)$ cannot be reduced.

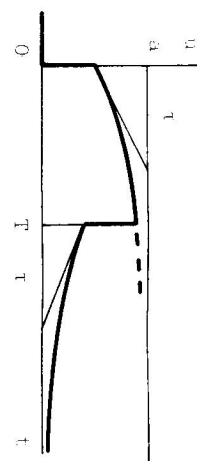
The conditioned controller, as given in figure 4b, ensures a good "piloting" of the state variable $v(t)$ and accordingly of the controlling signal $u(t)$; in this way the control loop is again in a normal situation as soon as the disturbance disappears.





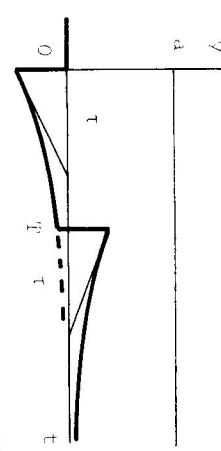
$$a(1 - e^{-\frac{t}{T}})$$

$$V_0 e^{-\frac{t}{T}}$$



$$a(1 - \frac{e^{-\frac{t}{T}}}{1+K})$$

$$V_0 \frac{e^{-\frac{t}{T}}}{1+K}$$

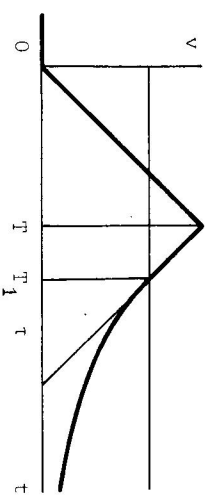


$$-a \frac{e^{-\frac{t}{T}}}{1+K}$$

$$V_0 \frac{e^{-\frac{t}{T}}}{1+K}$$

FIGURES 5A

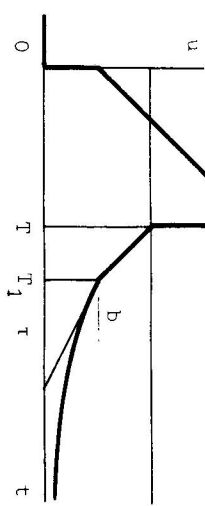
without any limitation



$$K(a-b) \frac{t}{T}$$

$$V_1 - Kb \frac{t-T}{T}$$

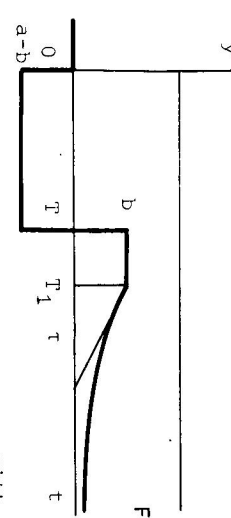
$$b(1+K) e^{-\frac{t-T}{T}}$$



$$K(a-b)(1 + \frac{t}{T})$$

$$V_1 - Kb(1 + \frac{t-T}{T})$$

$$b e^{-\frac{t-T}{T}}$$

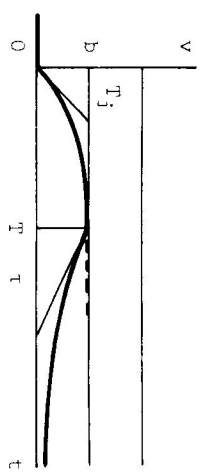


$$-(a-b)$$

$$b e^{-\frac{t-T}{T}}$$

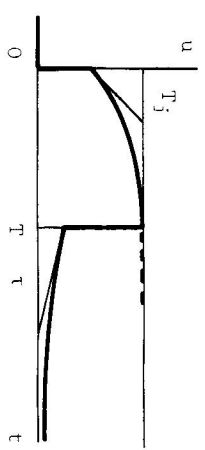
FIGURES 5B

with saturation and without any anti-windup system



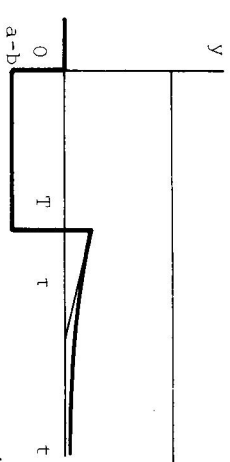
$$b(1 - e^{-\frac{t}{T}})$$

$$V_3 e^{-\frac{t}{T}}$$



$$K(a-b) + b(1 - e^{-\frac{t}{T}})$$

$$V_3 \frac{e^{-\frac{t}{T}}}{1+K}$$



$$-(a-b)$$

$$V_3 \frac{e^{-\frac{t}{T}}}{1+K}$$

FIGURES 5C

with saturation and with our anti-windup system

5. CONCLUSIONS

The solution of the windup problem has not to be different in each particular case.

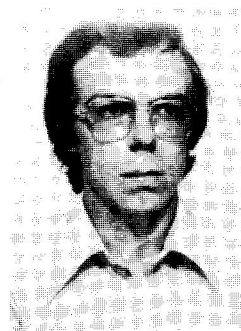
It is on the contrary possible to find a general method by which the disagreements of this phenomenon are suppressed or at least alleviated in a reasonable way.

The implementation of the proposed conditioning algorithm is in no way more complicated than that of the original unconditioned one. The results are good for any control structure. Moreover, the method is not limited to the one-input-one-output linear continuous case which is taken as an example in this paper. It can be easily extended to non-linear cases. The study of the multi-input-multi-output systems in connection with the windup problem, leads to foresee new control structures.

The case of the multi-input-multi-output systems will be addressed in a forthcoming paper.

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Raymond J.L. HANUS was born at Brussels, Belgium, on May 2, 1945. He received his degrees of physical engineering and of automatic control engineering from the Free University of Brussels in 1970 and 1972 respectively.

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